

Argus System Review (IPO)

3.1 Theoretical Basis for Behavioral Cue Analysis

ARGUS is based on the idea that a person's observable behavior during an online examination can provide useful information when examining potentially suspicious examination behavior. The system focuses on behaviors that can be captured through a webcam, particularly head and facial orientation, eye gaze, and upper-torso or posture-related movements. These behaviors are treated as cues that may contribute to the analysis of examination behavior, rather than as direct proof of academic dishonesty.

The theoretical foundation of ARGUS is supported by Interpersonal Deception Theory (IDT) and the Component Process Model (CPM). IDT provides a basis for understanding how deceptive situations may be associated with changes in observable nonverbal behavior. In an examination setting, changes in gaze, head movement, posture, or other nonverbal actions may provide information that can be considered when identifying behavior that may require further attention. However, these behaviors can also occur during normal examination activities and are not exclusive to deception.

The Component Process Model provides another perspective by explaining how emotional responses can develop from a stimulus, the person's appraisal of that stimulus, and the resulting physiological, motor, and expressive responses. During an examination, for example, a difficult question or concern about performance may cause a participant to experience stress or other emotional responses. These responses may be reflected in changes in gaze, facial orientation, posture, or movement. Because of this, a behavioral change detected by ARGUS does not necessarily have one specific meaning or cause.

For this reason, ARGUS treats gaze deviation, head movement, facial orientation, and posture changes as behavioral cues rather than definite indicators of cheating. An individual action, such as looking away from the examination screen or adjusting one's posture, may be completely normal. Instead of treating a single action as suspicious, ARGUS considers how different cues occur together and how they change over time. For example, repeated gaze deviation accompanied by changes in head orientation and upper-torso movement may provide more context than any one of these behaviors on its own.

This distinction between observing a behavior and interpreting that behavior is important to the design of ARGUS. The feature-extraction components identify measurable visual behaviors, while the multimodal processing stage brings these observations together for further analysis. The system therefore does not assign a fixed meaning to a particular movement. Instead, the observed behavior is considered in relation to other cues, their frequency, and how they occur over a period of time.

The three behavioral modalities used by ARGUS provide different types of information. Head and facial orientation describes the direction and movement of the participant's head and face. Eye gaze provides information about the participant's visual direction and changes in where the participant appears to be looking. Upper-torso and posture features provide information about changes in body position and movement that may not be captured through facial or gaze observations alone. Combining these modalities allows the system to consider examination behavior from multiple observable aspects rather than relying on a single cue.

Overall, the theoretical foundation of ARGUS supports the use of observable behavioral patterns as part of multimodal behavioral analysis while recognizing that individual behaviors can have different possible

explanations. The system therefore focuses on identifying patterns that may indicate behavioral suspicion rather than treating them as confirmation of academic dishonesty. Any behavior identified by the system is intended to provide information for further assessment, with the final interpretation remaining subject to human review.

3.2 Multimodal Behavioral Representation

Human behavior is expressed through different types of observable actions rather than through a single signal. During an online examination, a participant's behavior may be reflected through changes in eye gaze, head and facial orientation, and body posture. ARGUS uses these observable behaviors as multimodal behavioral cues. Each modality provides a different view of the participant's actions, and combining them allows the system to form a broader representation of behavior during an examination.

The use of multiple behavioral cues is important because a single action can have different possible explanations. For example, looking away from the screen may indicate distraction, thinking, or interaction with something outside the camera's view. Similarly, a change in posture may simply result from discomfort or an attempt to sit more comfortably. Because of this, ARGUS does not treat one behavioral cue as sufficient to determine suspicious behavior. Instead, the system considers the relationship between different cues and how they occur over time.

3.2.1 Head and Facial Orientation

Head and facial orientation describes the direction and movement of the participant's head and face during the examination. Changes in orientation can provide information about where the participant is facing and whether their attention appears to remain directed toward the examination screen. For example, repeated head turns or noticeable changes in head direction may indicate that the participant is looking toward a different area of the environment.

Head and facial orientation is useful because it provides information that may complement eye gaze. A participant may change their eye direction without significantly moving their head, while in other situations both the eyes and head may move together. Looking at these behaviors together provides more context than relying on either one independently.

In ARGUS, head and facial orientation is therefore treated as one of the behavioral inputs used to describe the participant's activity. Changes in orientation are not automatically interpreted as cheating because they may also occur during normal examination behavior. Their relevance depends on how they appear alongside other behavioral cues and whether similar patterns occur repeatedly over time.

3.2.2 Eye Gaze

Eye gaze refers to the direction in which the participant appears to be looking. During an online examination, gaze can provide an indication of whether the participant's visual attention is directed toward

the examination screen or toward another area. Changes in gaze direction can therefore provide useful information when examining possible changes in attention or behavior.

Gaze behavior can have several possible explanations. A participant may look away from the screen while thinking about a question, becoming distracted, or responding to something in the surrounding environment. For this reason, gaze deviation should not be considered direct evidence of cheating. Its significance becomes more meaningful when it is observed together with other behaviors or occurs repeatedly within a period of time.

Within ARGUS, eye gaze provides a behavioral cue that complements head and facial orientation. While head orientation describes the direction of the participant's head and face, gaze provides additional information about the participant's visual direction. Considering both allows the system to capture changes in attention that may not be fully represented by either modality alone.

3.2.3 Upper-Torso and Posture

Upper-torso and posture refer to observable changes in the participant's body position and movement within the camera's view. These include changes in the position of the shoulders and torso, leaning, body movement, and other posture-related actions that can be observed from the upper portion of the participant's body.

Posture can provide additional context because some behaviors cannot be understood through facial or gaze information alone. For example, a participant may change their body position while looking away from the screen, or their upper-torso movement may occur at the same time as changes in head orientation. These combined observations can provide a more complete description of what is happening during a particular period of the examination.

Similar to the other modalities, posture changes do not have a fixed meaning. Participants naturally move, adjust their seating position, or change their posture during an examination. ARGUS therefore treats these movements as behavioral observations that can contribute to the overall analysis rather than as direct indicators of dishonest behavior.

3.2.4 Temporal Behavioral Patterns

Behavior during an examination is not limited to individual moments. A participant's actions may occur once, continue for a period of time, happen repeatedly, or appear together with other behavioral cues. Because of this, the temporal pattern of a behavior is an important part of how ARGUS represents examination behavior.

An isolated action may have little significance on its own. For example, briefly looking away from the screen may be a normal part of answering a difficult question. In contrast, repeated gaze deviations, recurring head movements, or a combination of gaze, head, and posture changes occurring within the same period may provide more behavioral context. Examining these patterns over time helps distinguish individual events from behaviors that are repeated or sustained.

ARGUS therefore considers behavioral cues in relation to their occurrence over time. The extracted observations from the different modalities are organized according to their corresponding periods in the examination so that behaviors from different modalities can be considered together. This temporal representation provides the basis for the later multimodal fusion and behavioral inference performed by the system.

Overall, the multimodal behavioral representation of ARGUS consists of head and facial orientation, eye gaze, upper-torso and posture, and the temporal patterns in which these behaviors occur. Each modality contributes different information, while their combined representation provides a broader view of the participant's observable behavior. This representation serves as the foundation for the next stage of the system, where the behavioral cues are aligned and combined for multimodal behavioral analysis.

3.3 Multimodal Fusion and Behavioral Inference

The behavioral cues described in the previous section provide different types of information about a participant's actions during an online examination. However, examining each cue separately can make it difficult to understand the overall behavior. A change in eye gaze, for example, may have a different meaning depending on whether the participant's head and body also change at the same time. Multimodal fusion addresses this by bringing observations from different behavioral modalities together so they can be considered as part of the same behavioral context.

Multimodal fusion refers to the process of combining information from multiple sources to create a more complete representation of an observed behavior. In ARGUS, the sources are the behavioral observations obtained from head and facial orientation, eye gaze, and upper-torso or posture analysis. Each modality contributes information that may not be available from the others. Combining them allows the system to consider relationships between different behaviors instead of relying on a single observation.

This approach is important because individual behavioral cues can be ambiguous. For example, looking away from the examination screen may occur because a participant is thinking, distracted, or interacting with something in the surrounding environment. A posture change can also happen naturally during a long examination. However, when different cues occur together or repeatedly over time, they can provide additional context about the participant's behavior. Multimodal analysis therefore allows the system to consider the overall pattern rather than assigning a fixed meaning to one action.

In ARGUS, the relationship between gaze, head orientation, and posture is particularly important. These modalities can describe related aspects of the same behavior from different perspectives. For instance, a change in gaze may occur together with a corresponding head movement, while an upper-torso movement may occur during the same period. Considering these observations together provides a more complete behavioral representation than analyzing each modality independently.

The combined behavioral observations are processed through the Multimodal Behavior Inference Engine (MBIE), which serves as the central component for multimodal behavioral analysis in ARGUS. MBIE brings the observations from the different modalities together and organizes them into a structured multimodal behavioral representation. This representation allows behaviors from different sources to be considered in relation to one another and across the period in which they occur.

The purpose of MBIE is not to assign a fixed interpretation to every individual observation. Instead, it provides a basis for examining how different behavioral cues are related. The system can therefore consider whether a behavior is isolated, repeated, sustained, or accompanied by other observable cues. This helps maintain the distinction between detecting a behavior and determining what that behavior may mean.

After the behavioral observations have been combined, they can be used for behavioral inference. In this context, behavioral inference refers to the process of examining the available behavioral representation to identify patterns that may be relevant to suspicious examination behavior. The inference stage considers the combined observations rather than treating a single cue as a definitive indication of cheating.

This multimodal approach reflects the overall purpose of ARGUS as a behavioral suspicion classification system. The system uses observable behavioral patterns to identify situations that may require further attention, but the resulting interpretation does not establish that academic dishonesty has occurred. The combined behavioral representation instead serves as information that can support the later behavioral analysis and human review stages of the system.

3.4 ARGUS Conceptual Framework

3.5 Behavioral Risk Report and Decision Support

The final output of ARGUS is presented through a Behavioral Risk Report, which brings together the results of the behavioral analysis into a form that can be reviewed more easily. Rather than requiring a reviewer to examine individual behavioral observations separately, the report provides a summary of the detected behaviors and the system's interpretation of the observed patterns.

The report may include detected behavioral events, their corresponding timestamps, and the behavioral cues associated with those events. These details provide context about when a particular behavior occurred during the examination and allow the reviewer to relate the reported behavior to the corresponding portion of the recorded video. This is important because the same behavior can have different meanings depending on what was happening at the time it was observed.

The report also includes the interpretation produced during the LLM-based behavioral analysis stage. The LLM considers the structured behavioral information provided by the earlier stages of ARGUS and generates a natural language interpretation of the observed patterns. This allows the results of the multimodal analysis to be presented in a form that is easier for a human reviewer to understand rather than relying only on raw measurements or individual behavioral events.

As part of this interpretation, the system can identify whether the observed behavioral pattern is considered suspicious or non-suspicious based on the information provided to the analysis stage. The report can also provide an associated risk or likelihood indication and a natural-language explanation of the factors that contributed to the interpretation. These outputs are intended to summarize the system's assessment of the observed behavior rather than serve as definitive evidence of academic dishonesty.

The Behavioral Risk Report therefore serves as a decision support tool for human reviewers. A reviewer can use the reported events, timestamps, and behavioral interpretation to determine which portions of an examination may require closer examination. The reviewer can also compare the reported behavior with the original examination recording and consider possible explanations that may not be available to the system.

This human review component is important because behavioral observations are not always unambiguous. A participant may look away from the screen, change posture, or move their head for many legitimate reasons. The system can identify and organize these observable behaviors, but it cannot determine the participant's actual intent from the visual cues alone. The report should therefore be understood as a means of highlighting potentially relevant behavior rather than making a final judgment about whether cheating occurred.

Overall, the Behavioral Risk Report connects the technical analysis performed by ARGUS with the person responsible for reviewing the examination. By presenting behavioral events, their timing, and the system's interpretation in a structured report, ARGUS provides information that can help guide further human assessment while keeping the final decision outside the automated system.

3.6 Role and Boundaries of ARGUS

ARGUS is developed as a proof of concept system for identifying behavioral patterns that may be considered suspicious during an online examination. The system analyzes observable behaviors from recorded webcam videos, particularly head and facial orientation, eye gaze, and upper-torso or posture movements. These different cues are processed and combined to provide a broader view of the participant's behavior during an examination.

However, ARGUS is not designed to determine whether a participant actually cheated. Its purpose is to identify behaviors that may be worth looking into further. When the system detects a potentially suspicious behavior, the corresponding segment or timestamp can be included in the Behavioral Risk Report for review. In this sense, the output of ARGUS should be treated as an indication for further investigation rather than as proof of academic dishonesty.

This distinction is important because the behaviors monitored by ARGUS are not necessarily signs of cheating on their own. Looking away from the screen, turning the head, changing posture, or moving the upper body may occur naturally during an examination. For example, a student may look away while thinking about a question, adjust their sitting position, or briefly move outside the camera's ideal view. Because of this, a single behavioral cue should not be treated as enough evidence to conclude that suspicious activity has occurred.

Instead, ARGUS considers behavioral patterns across different cues and over time. The system combines the extracted information from head and facial orientation, eye gaze, and upper-torso movement through the Multimodal Behavior Inference Engine (MBIE). The combined information is then organized into a structured behavioral representation that can be interpreted by the Large Language Model (LLM). This allows the system to consider several observations together rather than relying only on an isolated movement or frame.

The Behavioral Risk Report generated by ARGUS is intended to support human review. It provides information about the detected behavioral events, including their timestamps and the evidence associated with them. This can help a reviewer locate specific parts of an examination recording that may require closer attention. The system therefore serves as a tool for assisting the reviewer rather than replacing human judgment.

The limitations of ARGUS are also important to consider. The system only works with the visual information available from the recorded webcam video. Behaviors that occur outside the camera's view, as well as other forms of possible academic dishonesty that cannot be observed through the available visual data, cannot be reliably identified by the system. In addition, the suspicious and non-suspicious labels used in this study are based on the scripted conditions of the experiment. They should therefore not be treated as independently verified instances of actual cheating.

Overall, ARGUS is intended to function as a decision-support system for behavioral suspicion classification. It can extract and combine observable behavioral cues and identify segments that may deserve human attention, but it does not independently establish that academic dishonesty occurred. Any final judgment should remain with a human reviewer who can consider the behavioral evidence together with the broader context of the examination.

4.1 Research Design

This study used a simulation based experimental design to develop and evaluate ARGUS as a proof of concept system for analyzing behavioral cues during online examinations. The evaluation was conducted offline using recorded mock examination sessions rather than through a live online proctoring setup. This allowed the researchers to process the same recordings through the different stages of the ARGUS pipeline and examine how behavioral observations were extracted, combined, and analyzed.

The experiment was conducted in a controlled setting where participants performed simulated examination activities under two scripted conditions: a simulated cheating condition and a simulated non cheating condition. The simulated cheating condition represented behaviors intended to resemble suspicious examination activity, while the simulated non cheating condition included normal examination behavior and actions that could potentially appear suspicious even when no cheating was intended. Using both conditions allowed the study to examine how ARGUS responded to different types of observable behavior.

The completed dataset consisted of eleven recorded mock examination sessions. Six sessions were assigned to the simulated cheating condition and five sessions were assigned to the simulated non cheating condition. The recordings were divided into 154 one-minute or final partial-minute processing segments. Of these segments, 81 came from sessions assigned to the simulated cheating condition and 73 came from sessions assigned to the simulated non cheating condition. Each segment inherited the scripted condition label of the session from which it came.

Three levels of experimental units were considered in the study: the participant, the recording or session, and the processing segment. The participant represents the person who performed the mock examination. A recording or session represents one complete mock-examination video under an assigned experimental condition. A segment represents a portion of a recording that was processed by the feature extraction and multimodal analysis pipeline. Segment level data were used to generate and examine

behavioral features, while the main classification evaluation was performed at the session level. This distinction was important because multiple segments came from the same recording and therefore could not be treated as independent participants.

The reference labels used in the evaluation represent the scripted condition assigned to each session. They should not be interpreted as independently verified labels indicating that cheating actually occurred at a particular point in a recording. Since every segment inherited the label of its source session, the 154 segments were not independently annotated for the occurrence, type, or exact timing of suspicious behavior. The study also included limited qualitative checking of extracted behavioral observations, but comprehensive multimodal behavioral annotation was not completed.

The overall research workflow followed the actual ARGUS processing pipeline. Recorded examination videos were first prepared and divided into processing segments. Each segment was then processed by the behavioral feature-extraction modules for head and facial orientation, eye gaze, and upper-torso or posture related behavior. The resulting observations were stored as structured data and aligned for multimodal processing. The Multimodal Behavior Inference Engine (MBIE) then combined the extracted behavioral information into a structured representation. This representation was subsequently provided to the LLM-based reasoning stage, which generated an interpretation and contributed to the final Behavioral Risk Report.

The study therefore evaluated the system at several stages rather than treating the final classification as the only outcome. The feature extraction stage was examined to determine whether the recorded videos could be converted into usable behavioral measurements. The multimodal stage was examined to determine whether observations from the different modalities could be aligned and represented together. Finally, the conventional baseline and LLM based reasoning outputs were evaluated at the session level to examine their ability to distinguish between the two scripted experimental conditions.

Because the experiment used simulated behaviors in a controlled environment, the results are intended as preliminary proof of concept evidence. The design allowed the researchers to examine the operation of the ARGUS pipeline under controlled conditions, but it does not reproduce all of the variation that may occur during authentic online examinations. In addition, the scripted condition labels describe the experimental setup and should not be treated as direct evidence of academic dishonesty. The purpose of the evaluation was to determine how the implemented system processes and interprets observable behavioral patterns, with its outputs intended to support further human review rather than independently establish that cheating occurred.

4.2 Dataset Construction and Experimental Setup

The dataset used in this study was created from controlled mock examination recordings designed to provide video data for the ARGUS behavioral analysis pipeline. The recordings were intended to represent both potentially suspicious examination behavior and ordinary examination behavior. This allowed the system to be tested under controlled conditions while also including behaviors that could naturally occur during an examination and potentially be mistaken for suspicious activity.

4.2.1 Participants and Recording Setup

The completed study consisted of eleven recorded mock examination sessions. These sessions were not equivalent to eleven unique participants because some participants contributed recordings under more than one scripted condition. This distinction is important when interpreting the evaluation because recordings from the same participant may share individual characteristics such as movement patterns, posture, and gaze behavior.

The recordings were collected using a single ASUS TUF Gaming F15 laptop and its built-in webcam. Videos were recorded at a resolution of 1280×720 pixels and stored in MP4 format. The webcam was positioned in a fixed frontal arrangement, with the laptop placed on a desk and the camera centered above the display at approximately eye level. This setup provided a view of the participant's face and upper torso, which were the main areas required by the feature extraction modules. Environmental conditions, including lighting and background, were kept as consistent as reasonably possible across the recording sessions.

The recorded videos varied in their actual capture rates and durations. The capture rate ranged from approximately 15 to 30 frames per second, while recording durations ranged from 4 minutes and 6 seconds to 18 minutes and 42 seconds, with an average duration of approximately 13 minutes and 22 seconds. The variation in recording length and capture rate reflects the actual collected dataset rather than a fixed recording duration imposed on every session.

Participants completed the same general mock-examination setup while following the assigned behavioral condition for the session. The examination environment was designed to resemble an online examination in which the participant remained in front of the computer and was recorded through the webcam. The mock examination materials used for the sessions are documented in the study's supplementary materials.

4.2.3 Reference Labels and Qualitative Review

Each recorded session was assigned a reference label according to its predetermined scripted condition: simulated cheating or non-cheating. The labeling was therefore performed at the session level. The 154 processing segments generated from the recordings inherited the label of their corresponding session.

These labels should not be interpreted as independently verified ground truth for cheating or suspicious behavior. A segment labeled as simulated cheating indicates that the recording came from a session assigned to that experimental condition; it does not mean that a specific cheating event was independently confirmed at that exact timestamp. Similarly, a non-cheating label represents the assigned experimental condition rather than proof that no suspicious-looking behavior occurred within the segment.

The researchers also conducted limited manual review of selected recordings to examine the relationship between reported system events and visible behavior. ELAN 7.0 was initially considered for comprehensive time aligned annotation of the behavioral modalities. However, complete annotation of gaze, head orientation, hand activity, and upper-torso behavior across the entire dataset was not completed because of the amount of frame-by-frame manual work required.

Instead, ELAN was used for limited qualitative spot checking. A single EyeGaze annotation tier was created using the categories CENTER, LEFT, RIGHT, UP, and DOWN. Selected system generated timestamps were then compared with the corresponding portions of the recordings to determine whether

the reported gaze behavior was visually plausible. This review was not intended to provide comprehensive event level ground truth or formal validation of the feature extraction modules.

4.2.4 Video Segmentation

After the recordings were collected, each video was divided into processing segments before being passed through the behavioral feature extraction pipeline. The purpose of segmentation was to divide longer recordings into manageable time intervals while preserving the timing information needed to relate extracted behaviors back to the original video.

The completed dataset produced 154 processing segments. Most segments represented one minute of video, while the final segment of a recording could be shorter when the remaining video duration was less than one minute. The resulting dataset contained 81 segments from sessions assigned to the simulated cheating condition and 73 segments from sessions assigned to the simulated non-cheating condition.

Each segment retained information linking it to its source recording and its position within the original video. This allowed the feature extraction modules to produce timestamped observations while maintaining the connection between individual behavioral measurements and the corresponding examination session.

The segments served as the primary processing units for the feature extraction stage. Head and facial orientation, eye gaze, and upper-torso or posture observations were generated from the corresponding video intervals and stored as structured outputs. These outputs were later aligned and combined during multimodal processing. Although the behavioral features were generated at the segment level, the main classification evaluation was conducted at the session level so that multiple segments from the same recording were not treated as independent participants.

Overall, the dataset construction process produced a controlled collection of mock examination recordings that could be processed consistently by ARGUS. At the same time, the use of scripted conditions, a single recording setup, and a limited number of sessions means that the resulting dataset represents a development and proof of concept corpus rather than a representative sample of real online examinations.

4.3 System Architecture and Processing Pipeline

The implemented ARGUS-Lite system follows a sequential processing pipeline that converts recorded examination video into structured behavioral data for multimodal analysis. The pipeline was designed around three visual behavior modalities: head and facial orientation, eye gaze, and upper-torso posture and movement. Rather than sending the raw video directly to the language model, the system first extracts measurable behavioral features from the video and stores them as timestamped JSON data. These outputs are then passed to the Multimodal Behavior Inference Engine (MBIE), where the different behavioral streams are aligned and combined for further analysis.

The implemented workflow can be summarized as:

Recorded Examination Video → Video Preprocessing and Segmentation → Parallel Feature Extraction → Structured JSON Outputs → MBIE → Behavioral Analysis and Risk Report

The processing stages are separated into independent feature-extraction modules so that each behavioral modality can be processed using a method appropriate to the type of visual information being measured. All three modules operate on the same segmented examination recordings and preserve timestamps so that their outputs can later be aligned by MBIE. The extraction modules use a nominal processing target of 15 frames per second, while the original recordings themselves were captured at varying frame rates.

4.3.1 Video Input and Preprocessing

The pipeline begins with the recorded mock-examination videos collected using the standardized recording setup described in Section 4.2. The recordings were captured using the built-in webcam of an ASUS TUF Gaming F15 laptop at 1280 × 720 resolution and stored in MP4 format. The camera was positioned directly in front of the participant, with the webcam centered above the display at approximately eye level. This framing allowed the system to observe the participant's face and upper torso throughout the examination session.

Before feature extraction, each recording is divided into approximately 60 second processing segments. A final segment may be shorter when the total recording duration does not divide evenly into one minute intervals. Each segment retains information identifying its source video, segment identifier, start and end timestamps, and target processing frame rate. This allows outputs from the individual feature extraction modules to remain traceable to their original location in the examination recording.

The segmented videos are then processed by the three feature-extraction modules. The modules operate independently on the same video intervals and generate timestamped observations rather than immediately making a final classification. This separation is important because the purpose of preprocessing is to transform visual information into structured behavioral measurements that can be combined later by MBIE.

Each module produces structured JSON output containing the extracted measurements and relevant processing metadata. The outputs preserve the temporal information of the original video, allowing observations from different modalities to be associated with the same point or interval in the examination session.

4.3.2 Head and Facial Orientation Feature Extraction

Head and facial orientation is extracted using MediaPipe FaceMesh together with an OpenCV Perspective-n-Point (PnP) solver. MediaPipe FaceMesh provides the facial landmarks required to estimate the position and orientation of the participant's face. A six-point facial model is then used to establish correspondences between detected two-dimensional facial landmarks and predefined three-dimensional facial reference points. These correspondences are processed using the PnP method to estimate the participant's head orientation.

The resulting orientation is represented using Euler angles for yaw, pitch, and roll. These measurements provide information about how the participant's head is positioned relative to the camera. For example,

yaw represents side-to-side head rotation, while pitch and roll describe other changes in head orientation. The system also records the participant's face position within the frame, providing additional spatial information that can be used during multimodal analysis.

To reduce frame-to-frame fluctuations caused by small landmark movements, the estimated Euler angles are smoothed using an exponential moving average with a smoothing factor of 0.4. This allows the system to retain meaningful head movements while reducing minor detection noise.

The module also measures mouth activity using the relative distance between selected mouth landmarks. A normalized mouth-aperture threshold of 0.03 is used to identify noticeable mouth movement. In addition, an off-screen head-orientation flag is generated when the absolute yaw exceeds 25 degrees. This threshold was selected during pilot testing to capture substantial head turns while allowing normal head movement associated with reading and examination activity.

The module processes each sampled frame independently and writes the resulting measurements directly to the corresponding segment output. Each observation is timestamped, allowing head-orientation events to be aligned with gaze and posture observations during later fusion. When the facial landmarks cannot be reliably detected or the pose calculation fails, the corresponding measurement is recorded as unavailable rather than being treated as a valid behavioral observation.

4.3.3 Eye-Gaze Feature Extraction

Eye-gaze estimation uses the iris landmarks provided by MediaPipe FaceMesh with landmark refinement enabled. The system estimates horizontal gaze direction by locating the center of each iris and comparing it with the center of the corresponding eye region. The left and right iris centers are calculated from their respective iris landmark sets, while the eye boundaries are obtained from facial landmarks surrounding each eye.

The horizontal displacement of the iris relative to the eye region is normalized by eye width and converted into an angular measurement. The resulting gaze angle provides a continuous estimate of how far the participant's gaze has moved from the defined center reference. The system also retains separate measurements for the left and right eyes, together with the calculated eye ratios used during the estimation.

The estimated angle is converted into categorical gaze regions using two thresholds. A gaze within 7 degrees of the center is classified as center, while angles beyond this range are classified toward the left or right. When the absolute gaze angle exceeds 12 degrees, the system marks the gaze as offscreen. These thresholds were empirically selected and are treated as configurable parameters rather than universal measures of suspicious behavior.

The gaze module focuses on horizontal gaze. Vertical gaze is not estimated because the implemented geometric approach does not provide a sufficiently reliable vertical gaze measurement without a more complex three dimensional eye model. This limitation is important when interpreting gaze related results because looking upward or downward cannot be represented with the same angular measure used for horizontal gaze.

Unlike a simple frame level flag, the gaze module also maintains a cumulative off-screen duration across consecutive off-screen observations. This allows the generated output to retain information about sustained gaze deviation rather than only recording whether a single frame exceeded the threshold. The resulting JSON contains timestamped gaze observations, continuous gaze angles, categorical gaze regions, detection status, and off-screen duration information.

The gaze features are written to the segment output at the target sampling rate without removing the individual frame level observations. Preserving these observations gives MBIE access to the temporal sequence of gaze behavior and allows it to determine whether gaze deviations occur briefly, repeatedly, or for sustained periods.

4.3.4 Upper-Torso and Posture Feature Extraction

Upper-torso and movement features are extracted using MediaPipe Pose. The implemented configuration uses the pose model to obtain body landmarks that describe the participant's upper-body position and movement. From these landmarks, the system derives several behavioral indicators and continuous measurements related to posture, shoulder movement, arm position, and hand location.

One of the implemented indicators is hand-near-mouth proximity, which uses a normalized distance threshold of 0.20 to identify when a hand is positioned close to the mouth. The module also evaluates changes in upper-torso posture using relationships among the shoulders, hips, and torso center. Changes in shoulder tilt, hip tilt, torso-center position, shoulder width, or hip width can contribute to a posture-shift indication.

Additional indicators include elevated hands and high movement. Elevated hands are identified by examining the position of the wrists relative to the shoulders, while movement is determined from changes in the torso center between consecutive sampled frames. The system also calculates continuous measurements such as torso angle, forward lean magnitude, and arm asymmetry. These measurements provide more detailed information than a simple binary suspicious/non-suspicious flag and allow the downstream fusion stage to examine the degree and persistence of a movement.

A torso center tracker with exponential smoothing is maintained across frames to provide a more stable reference point and reduce the effect of short term detection changes. The implemented forward lean measure should also be interpreted according to the actual system design: it represents lateral torso displacement relative to the horizontal screen center rather than a calibrated three-dimensional depth measurement.

As with the other extraction modules, posture observations are generated for each sampled frame and stored in the corresponding segment JSON. The output contains both the frame-level measurements and segment level summaries, including ratios for selected behavioral flags. This structure preserves the temporal information needed by MBIE while also providing compact summaries that can be used during subsequent behavioral analysis.

A limitation of this module is that MediaPipe Pose depends on sufficient visibility of the upper body. Occlusion of important landmarks, particularly around the shoulders or hips, can cause pose detection to fail. In such cases, the absence of a valid pose measurement cannot by itself be interpreted as a neutral

or suspicious behavior. The posture indicators are also threshold-based and were implemented as interpretable heuristics rather than learned behavioral classifiers.

Taken together, the three feature extraction modules transform the original examination video into three complementary streams of structured behavioral observations. The head/facial module describes face position, head orientation, and mouth activity; the gaze module describes horizontal visual attention and off-screen duration; and the posture module describes upper-torso position and movement. Because all three outputs retain timestamps and segment identifiers, they can be brought together in the next stage of the pipeline, where MBIE performs temporal alignment and multimodal fusion.